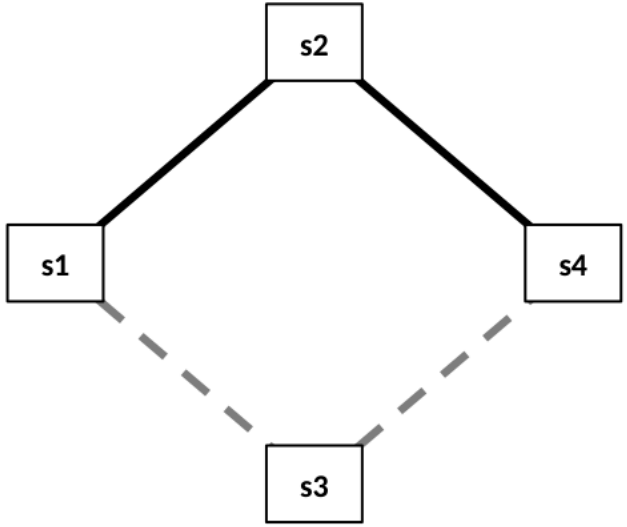
SmartTE

Centralized Traffic Engineering in Data Center Networks

BACSE203 Computer Networks, Fall 2026 | Topic 9

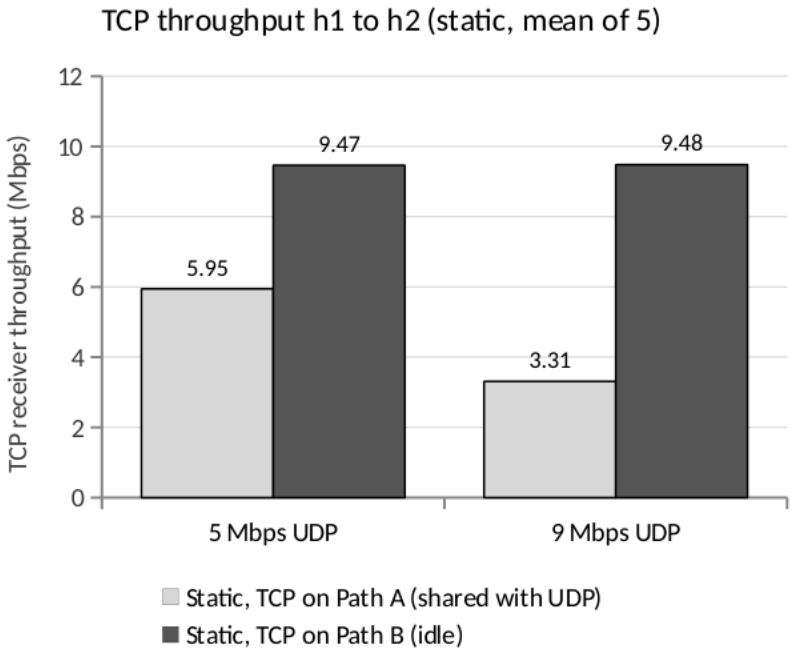
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Problem: static paths waste path diversity

- Data-center fabrics offer multiple paths, but static forwarding can leave a flow on a congested one
- SDN lets a controller read link counters and rewrite flow rules over OpenFlow
- Open question: does reacting to utilization help, and what does it cost in route changes?



Source: report Table 1 (E2 static contrast, n = 5).

Objectives and research question

Research question

When a controller scores paths by maximum link utilization, do a switching margin and a minimum dwell time reduce route oscillation, and what do they cost?

1

Build

OS-Ken controller (OpenFlow 1.3) that reads port counters and installs flow rules

2

Compare

Static, Policy B (lower utilization) and Policy C (margin + dwell)

3

Evaluate

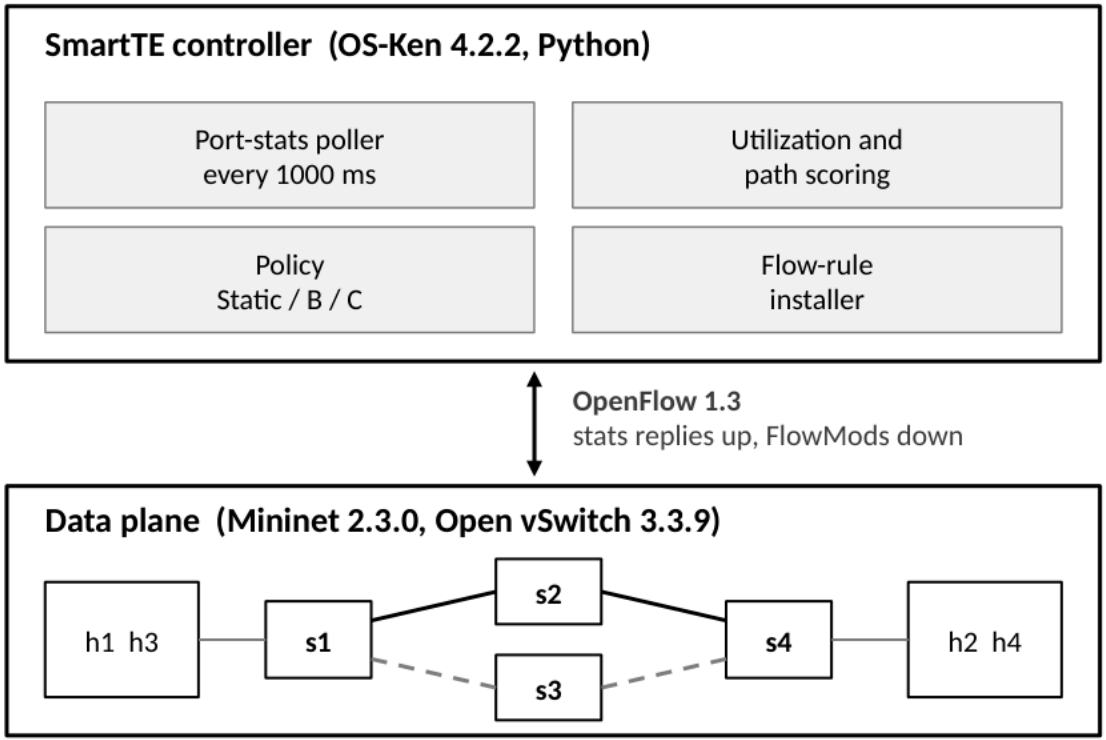
Four scenarios E1 to E4 in Mininet; throughput, RTT, loss, route events

4

Report

Trade-offs honestly; characterize this implementation, not production TE

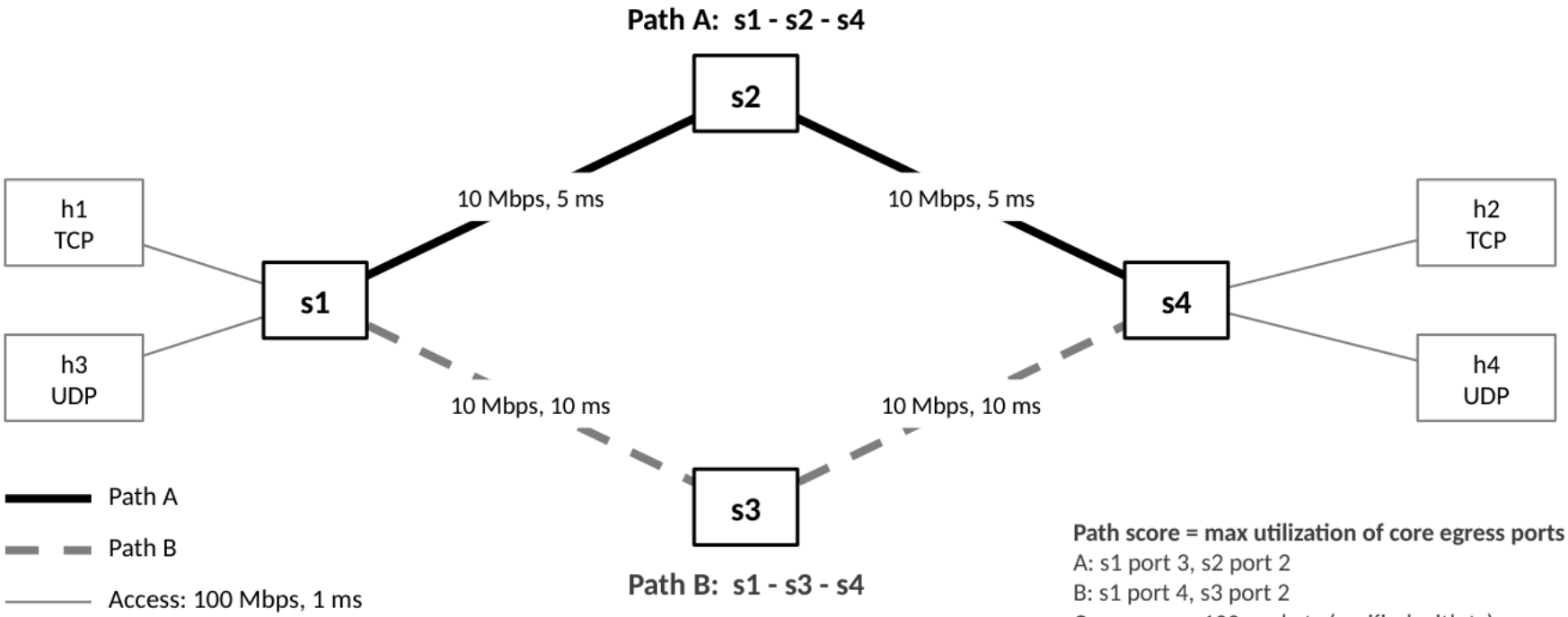
System architecture



Environment

- WSL2, Ubuntu 24.04.5 (aarch64)
- Mininet 2.3.0, OVS 3.3.9
- OS-Ken 4.2.2 (Python 3.10 env)
- iperf3 for TCP, ping for RTT/loss
- 20 unit tests pass (scoring, hysteresis, counters)

Test topology: four hosts, four switches



Path score = max utilization of core egress ports
A: s1 port 3, s2 port 2
B: s1 port 4, s3 port 2
Core queue: 100 packets (verified with tc)

Three forwarding policies compared

Static

baseline

Keeps the configured path. Never makes a routing decision.

–

0 route changes

Reference for what each path gives the TCP flow

Policy B

congestion-aware

Switches whenever the other path has a measurably lower utilization score.



16 route changes per trial (E1, no competitor)

Reacts fast (about 0.2 s in E2)

Policy C

SmartTE: hysteresis + dwell

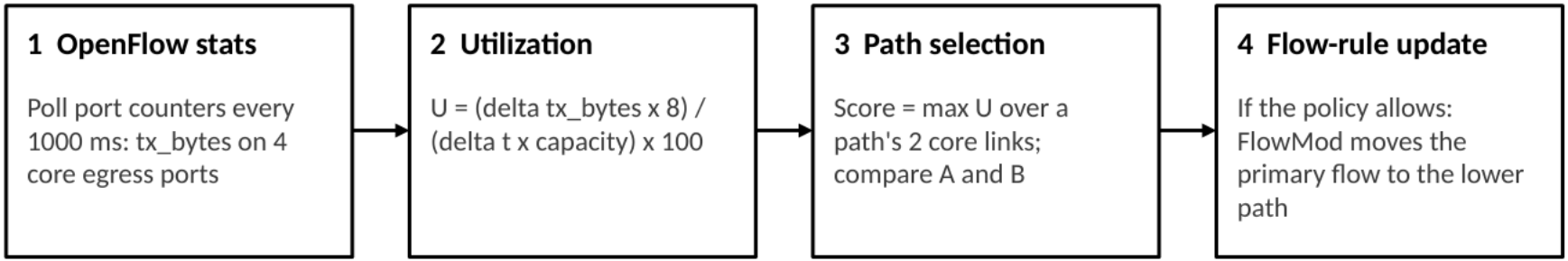
Also needs a 15 pp improvement and 5,000 ms since the last change.



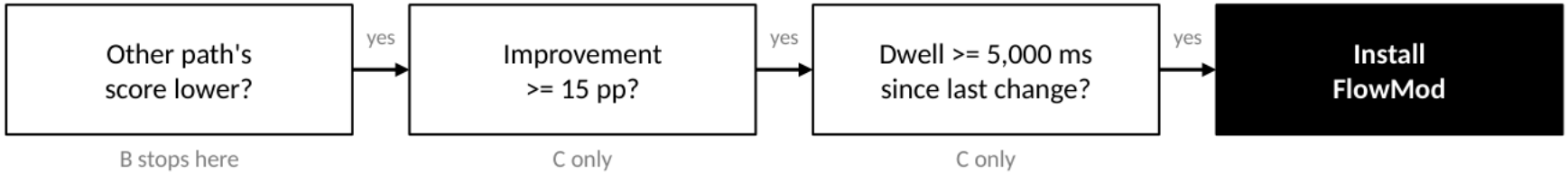
3 route changes per trial (E1, no competitor)

Reacts slower (about 2.2 s in E2)

Controller workflow

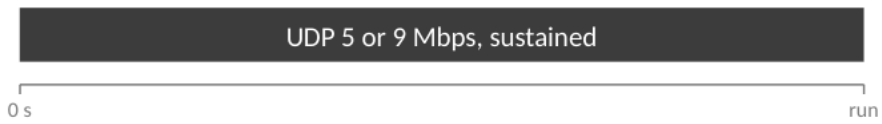
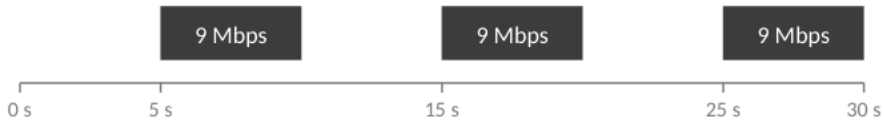


Policy gate (decision at step 4)



Any 'no' keeps the current path. Static never reaches the gate; B uses the first check only.

Experimental setup: scenarios E1 to E4

Scenario		UDP load on Path A over time
E1	**No competing load** TCP flow only (h1 to h2)	
E2	**Sustained congestion** UDP 5 or 9 Mbps on Path A; static A and B as references	
E3	**Burst schedule** Three 5 s bursts of 9 Mbps UDP in a 30 s run; C margin/dwell variants	
E4	**Path A link failure** Link s2-s4 disabled 1 s into a 15 s run	

Receiver-side iperf3 (TCP throughput), ping (RTT, loss), OpenFlow counters (utilization). 110 trials, n = 5 per cell, descriptive 95% intervals only.

Congestion results: throughput, RTT and packet loss (E2)

Mean [descriptive 95% CI]; five trials per condition. Ping loss is receiver-side ICMP loss.

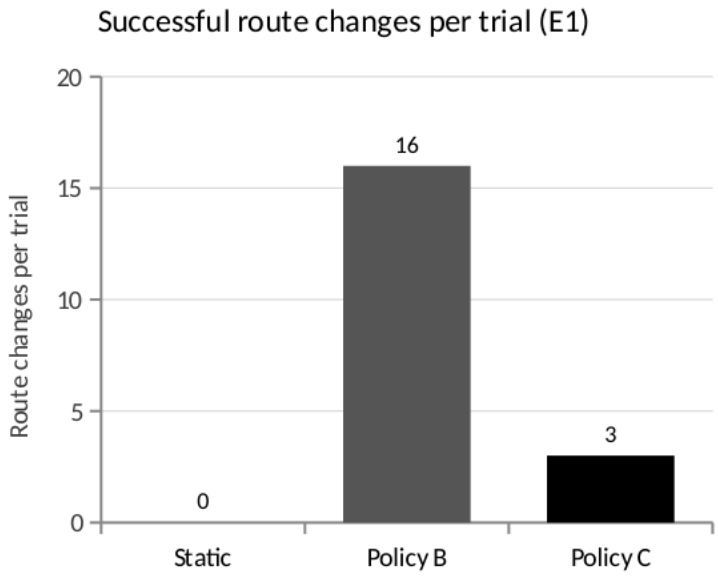
UDP load	Policy / path	TCP throughput (Mbps)	Mean ping RTT (ms)	Ping loss (%)
5 Mbps	**Static (A)**	5.95 [5.83, 6.06]	159.86 [145.37, 174.35]	3.40 [0.54, 6.26]
	Static (B)	9.46 [9.40, 9.53]	195.68 [169.36, 222.00]	1.60 [0.49, 2.71]
	Policy B	6.26 [5.96, 6.56]	51.80 [49.79, 53.81]	10.60 [3.99, 17.21]
	Policy C	7.44 [7.30, 7.58]	140.72 [136.63, 144.81]	4.40 [3.29, 5.51]
9 Mbps	**Static (A)**	3.31 [3.16, 3.46]	152.75 [142.46, 163.03]	5.20 [1.54, 8.86]
	Static (B)	9.48 [9.45, 9.52]	203.64 [185.71, 221.56]	1.20 [0.00, 2.82]
	Policy B	6.22 [5.76, 6.68]	68.31 [60.15, 76.47]	12.20 [7.69, 16.71]
	Policy C	9.45 [9.43, 9.47]	206.71 [187.90, 225.52]	1.40 [0.72, 2.08]

At 9 Mbps, Policy C preserved TCP throughput but had higher RTT; Policy B had lower RTT but higher ping loss.

Intervals are descriptive t-based 95% CIs (n = 5); ping-loss lower bounds are clipped at the physical limit of 0%.

Source: processed campaign m5_campaign_20261011T115819Z, E2-congestion metrics_summary.csv.

Route stability and switching behaviour



Across E1 to E3: B made about 14 to 25 changes per trial, C about 2 to 9 (report Fig. 4).

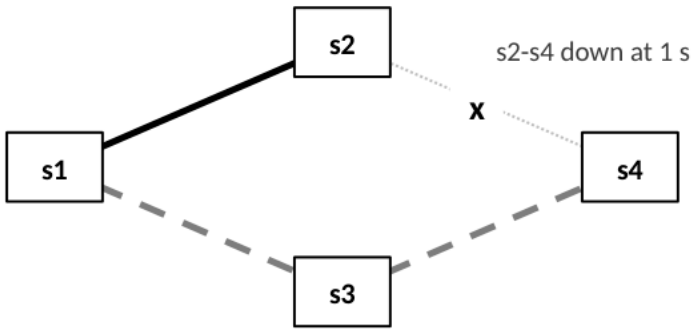
E3: Policy C sensitivity (margin / dwell)

Margin / dwell	Changes per 30 s trial	TCP (Mbps)
10 pp / 5 s	about 5	about 9.1
15 pp / 2 s	about 8	about 8.9
15 pp / 5 s (frozen)	**about 5**	**about 8.9**
15 pp / 10 s	about 3	about 7.9
30 pp / 5 s	about 5	about 8.9

Fewer changes can cost throughput: the largest dwell (10 s) switched least but lowered throughput. No best pair is claimed.

Link failure (E4): diagnostic, not a ranking

Policy	TCP results	On Path B at failure	Successful changes / trial	Attempts / rejected per trial	Recovery (s)	Ping loss (%)
Static	0/5	0/5	0	0 / 0	n/a	93.33
Policy B	5/5	5/5	1	17 / 16	0.193	2.13
Policy C	5/5	0/5	1	11.8 / 10.8	1.005	1.87



- B had already moved to Path B before the failure; C was still on A and switched after it
- So E4 cannot rank failover latency fairly
- Raw decisions are not changes: every trial had 1 successful change, the rest were rejected
- Static stayed on the failed path: no TCP result in 5/5 trials

Limitations and lessons learned

Limitations

- Single-host Mininet/OVS on WSL2: timing, Linux queues and a recorded HTB warning can affect results
- One TCP flow, one UDP competitor, four switches, two paths
- n = 5 per cell; descriptive intervals, no significance test
- Git state was dirty (git_dirty = true), so the exact source is not fully pinned

Lessons learned

- The controlled flow drives its own utilization signal: B switches even with no competitor
- Margin + dwell cut route changes but slow reaction (about 2.2 s vs 0.2 s)
- Count successful changes, not decisions: E4 raw 17 vs 1 successful
- Verify assumptions: queue limit of 100 packets checked with tc

Literature review and research gap

Prior work motivates the problem; SmartTE tests a narrower, implementation-specific question.

What prior work establishes

Centralized traffic engineering

B4 and SWAN coordinate paths with demand, rates and priorities at WAN scale [8, 9].

Data-center multipath

Hedera, MicroTE and CONGA use global scheduling, traffic patterns or distributed congestion signals [7, 11, 19].

Stability and safe updates

ECMP disruption, consistent updates and hybrid-routing instability are distinct concerns [2, 13, 14].

Research gap addressed here

Not a novelty claim

Centralized traffic engineering and route-stability mechanisms are established; this project does not claim otherwise.

Bounded experiment

Under one two-path topology and periodic max-link-utilization score, compare lower-score switching (B) with a 15 pp margin plus 5 s dwell (C).

What we measure

Successful route changes alongside TCP throughput, ping RTT/loss and controller observations across E1–E4.

Contribution: a controlled emulation of this feedback loop, not a direct benchmark against production systems.

Evidence boundary: 110 trials, n = 5 per cell; descriptive intervals. No universal superiority or statistical significance is claimed.

Comparison with representative research

Different scopes and evaluation methods: this is a conceptual comparison, not a numerical benchmark.

Work	Approach / evidence	Relation to SmartTE
OSPF / ECMP [1, 2, 15, 16]	IGP link-weight optimization or per-flow hashing; protocol analysis and network-wide studies.	Different control granularity; not this per-poll centralized congestion feedback.
Hedera / MicroTE / CONGA [7, 11, 19]	Large-flow scheduling, traffic-matrix prediction or distributed congestion feedback; data-center studies.	More global flow context or in-network coordination than SmartTE's one-flow, two-path setup.
B4 / SWAN [8, 9]	WAN-scale path, rate and priority coordination; production, prototype and data-driven evaluation.	Different scale and objectives; SmartTE does not allocate rates or claim comparable performance.
Reitblatt / Tseng [13, 14]	Consistent network updates and routing-stability methods; formal mechanisms and simulation.	Update correctness and stability are related but distinct; SmartTE measures a margin/dwell policy in emulation.
SmartTE (this study)	OS-Ken/OpenFlow 1.3; 1 s port-counter polling; Static vs B vs C in Mininet E1–E4.	Measures throughput, RTT, ping loss and successful route changes; bounded evidence (n=5 per cell).

Sources: report references [1], [2], [7]–[9], [11], [13]–[16], [19].

Conclusion

5.9 to 9.5

Mbps TCP, static on A vs B (5 Mbps UDP)

16 vs 3

route changes per trial, B vs C (E1)

110

trials, n = 5 per cell, 20/20 unit tests pass

- Path diversity helps only when assignment reacts to load
- A margin and dwell timer reduce route changes but delay action
- Evidence is specific to this emulation; no universal superiority is claimed

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